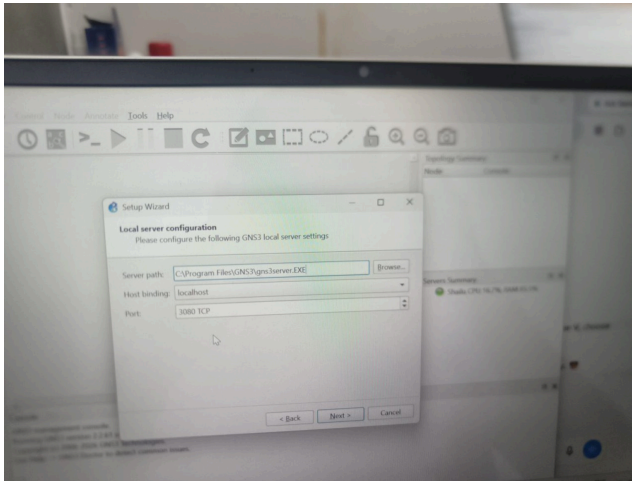


Prompts

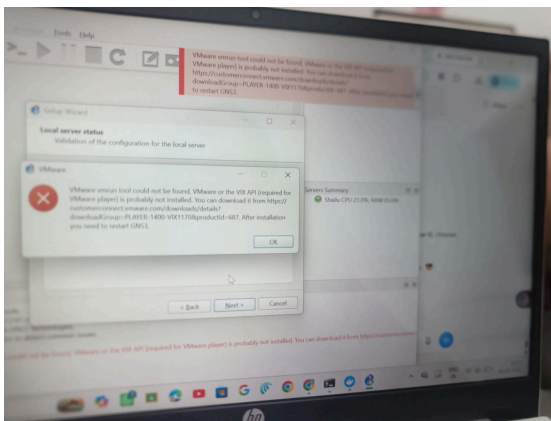
- 1) Brohh... naaku ee project problem statement ichharu...Develop a standardized framework to collect, transport and visualize network performance metrics like bandwidth usage and latency from routers and switches in real time.....First asalu ee problem ni simple gaa explain cheyyava.....Manam exact gaa em build cheyyali?
- 2) Brohhh problem statement ardham ayyindi konchem butnetwork telemetry framework..... ante exact gaa entiNormal network monitoring ki telemetry ki difference enti.....Oka simple example tho cheppu
- 3) Brohhh.....Project lo bandwidth usage, latency ani mention chesaru kadha.Routers/switches nunchi ee metrics actual gaa ela vastayi..... Device direct gaa bandwidth value istundha or maname calculate cheyyala?
- 4) Tools lo Python, gRPC ..SNMP... REST APIs... OpenTelemetry.... Grafana ani unnayi..... Brohhh..... okkokkati endhuku use chestham?.....Same project lo anni ela connect avuthayo simple flow lo cheppu.
- 5) Brohhh.....SNMP ante enti brohhh? Router ledha ...switch nunchi metrics collect cheyyadaniki SNMP endhuku use chestham..... SNMP device ki traffic generate chestundha leka just existing counters read chestundha?
- 6) Okay SNMP counters read chestundi ante, bandwidth calculate cheyyadaniki which counters kavali..... RX/TX bytes cumulative values aithe two polls madhya bandwidth ela calculate chestham?
- 7) Brrohh.....Latency kuda SNMP nunchi vastundha brohhh? Leka ping use cheyyala? Project requirement lo latency undhi kabatti correct approach decide cheyyali.
- 8) Broohhhh....Expected outcomes lo real-time dashboard.... threshold alerts...historical REST API...modular and scalable system unnayi..... Veetini functional requirements and non-functional requirements gaa separate chesi cheppu.
- 9) Brohhh RFC 9232 reference ichharu.....Mana project perspective lo RFC motham kakunda relevant concept enti..... Architecture design chestappudu em idea teesukovali
- 10)Nenu ila think chestunna brohh.....: Router next SNMP next Python Collector next Backend next Database next..... Grafana. Idi correct aa? Missing components emaina unnaya?
- 11)Collector nunchi backend ki direct REST use cheyyachu kadha brohhh? Mari project lo gRPC undhi..... . gRPC use chesthe advantage enti and exactly ekkada place cheyyali?

- 12) Okay brohhh, architecture roughly ardhham ayyindi. But actual system lo
Router/Switch Collector Backend Database
Grafana ani flow enduku Okkokka component responsibility simple
example tho explain cheyyi.
- 13) See brohhh.... Mana Network Telemetry Monitoring project ki early ga
end-to-end demonstrable version ready cheyyadaniki Minimum Viable
Implementation (MVP) ni define cheyyi..... . Core MVP lo YAML + .env
configuration, simulation mode lo realistic bandwidth/latency/loss/failure metrics
generate cheyyadam, counter-wrap-safe metric calculations, collector nunchi
backend ki gRPC client-server communication, FastAPI backend with
PostgreSQL storage and required GET APIs, threshold breach ki ACTIVE alert
and recovery ki RESOLVED alert logic, Grafana lo Overview, Bandwidth and
Alerts dashboards, and complete Docker Compose integration include cheyyi.
Real SNMP monitoring with physical/virtual devices, CSV export, full test suite,
SNMPv3 support, and OpenTelemetry tracing ni secondary features ga separate
cheyyi..... . Overall ga MVP lo first implement cheyyalsina core components
and later phases lo add cheyyagalige features ni clearly organize cheyyi.
- 14) Broooh, mana project ki custom simulator use cheyyadam better aa leka GNS3 lo
actual virtual router use cheyyadam better aa..... Both approaches
advantages/disadvantages compare chesi final development approach suggest
cheyyi.
- 15) One doubt brohhh: GNS3 lo VyOS router use chesthe adi actual router laga behave
chestundha..... SNMP enable chesi Python collector nunchi metrics collect
cheyyagalama?
- 16) Brooohh ardhham ayyindhi brohh Nextt build chedham brohh the
network-device side. GNS3.. VyOS setup ni complete gaa start nunchi guide
cheyyi. Every step verify chesaka next step ki vellali.
- 17) Broooooh, GNS3 install ayindi. Ippudu mana project kosam simple topology create
cheddham..... Router, switches and PCs ekkada use cheyyali..... Exact
topology diagram ivvu.
- 18) Broooooh, GNS3 install ayindi and ippudu mana existing Network Telemetry
project lo simulated telemetry source badulu GNS3/VyOS ni real-device validation
kosam use cheyyali..... . Mana project requirements ni consider chesi simple
GNS3 topology suggest cheyyi..... Which VyOS routers and switches use
cheyyali, avi exact ga ela connect cheyyali, and later mana Python collector

nunchi SNMP and ICMP tho ee devices ni monitor cheyyadaniki topology ela undali ani explain cheyyi. Exact topology gurincheyy discuss cheddham.

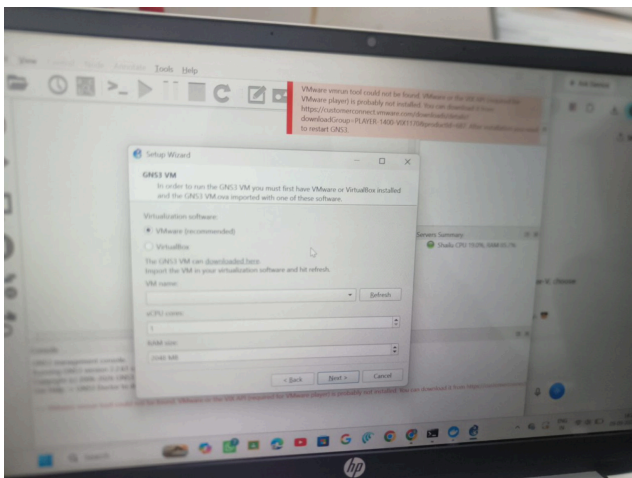


19)

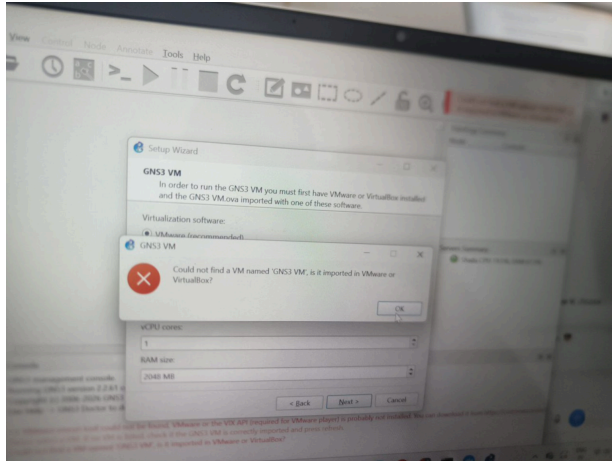


20)

Broooh, VyOS template create chesanu but node start chesthunte boot/QEMU error vastundi. Ee screenshot chusi problem ISO related aa, QEMU configuration aa, leka virtualization issue aa identify cheyyi. Mana current GNS3 VM setup ki safest fix cheppu.



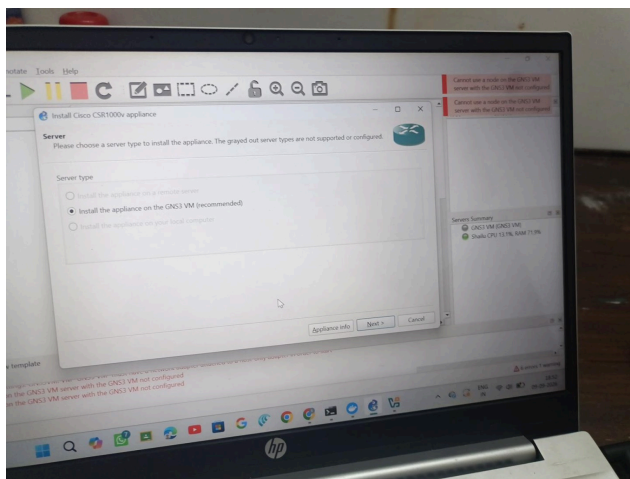
21)



22)

Brooooooh, GNS3 setup chesthunte ee error vachindi. Screenshot ni carefully chusi actual problem enti ani identify cheyyi. Mana existing GNS3 VirtualBox setup ni consider chesi, already imported GNS3 VM ni disturb cheyyakunda exact ga em check/change cheyyalo step-by-step cheppu. Random troubleshooting or unnecessary reinstallation suggest cheyyaku. BrrohhhhFix this erroorr brohhh

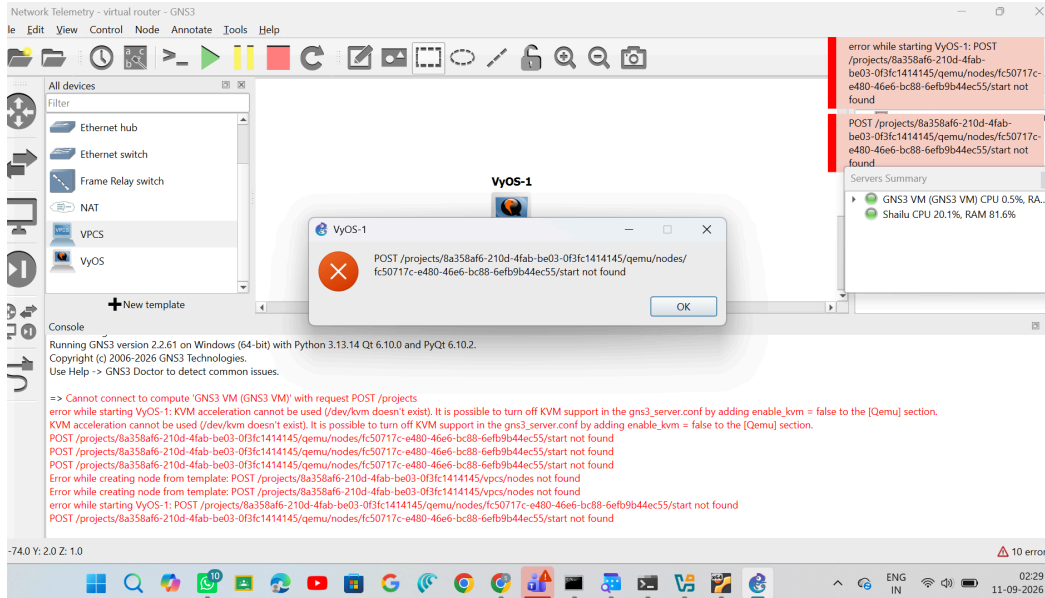
23) Bro, Cisco CSR1000v appliance install chesthunte ee screen vachindi, but right side lo “Cannot use a node on the GNS3 VM server with the GNS3 VM not configured” ani error kuda undi. Screenshot ni analyze chesi actual issue enti, GNS3 VM configuration lo exactly em missing undo identify cheyyi. Mana existing VirtualBox ... GNS3 VM setup ni consider chesi, already configured setup ni disturb cheyyakunda exact next steps cheppu.



24)

25) GNS3 lo devices add chesanu but links connect cheyyadam lo confusion undi. PC and router interfaces ni correct gaa ela connect cheyyali? Ethernet0/eth0 difference kuda cheppu.

26)

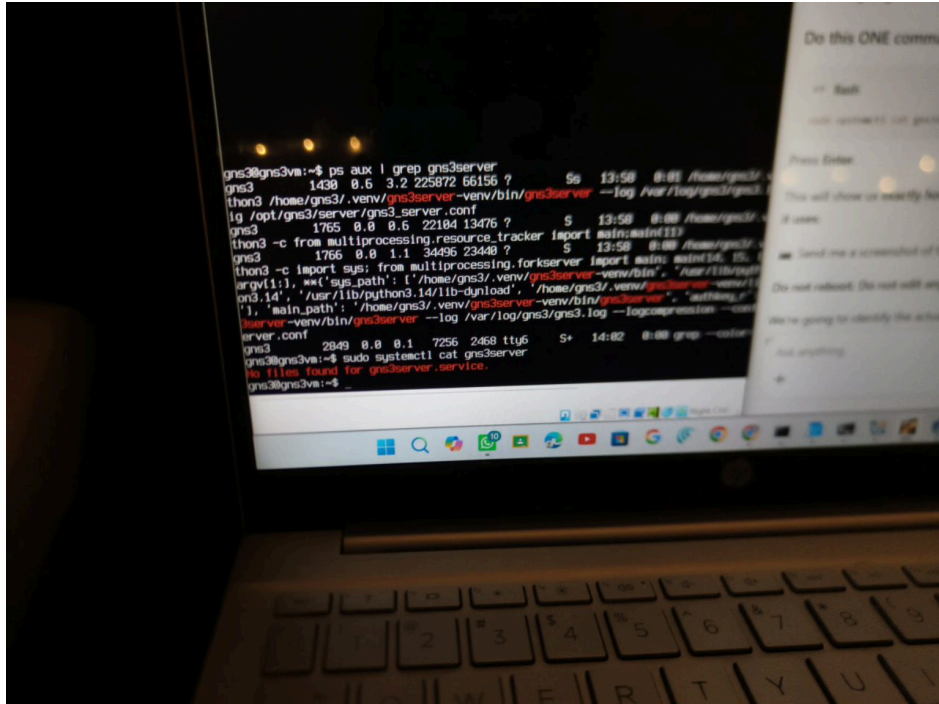


Broooh, GNS3 lo VyOS-1 start cheyyadaniki try chesthunte ee error vastundhi. Screenshot ni analyze chesi exact problem enti, enduku vastundho explain cheyyi and step-by-step resolution cheppu. GNS3 VM server green ga connected undhi.

Nenu Windows lo GNS3 2.2.61 use chesthunna, GNS3 VM kuda running lo undhi. VyOS QEMU appliance configure chesanu but start avvattledu.

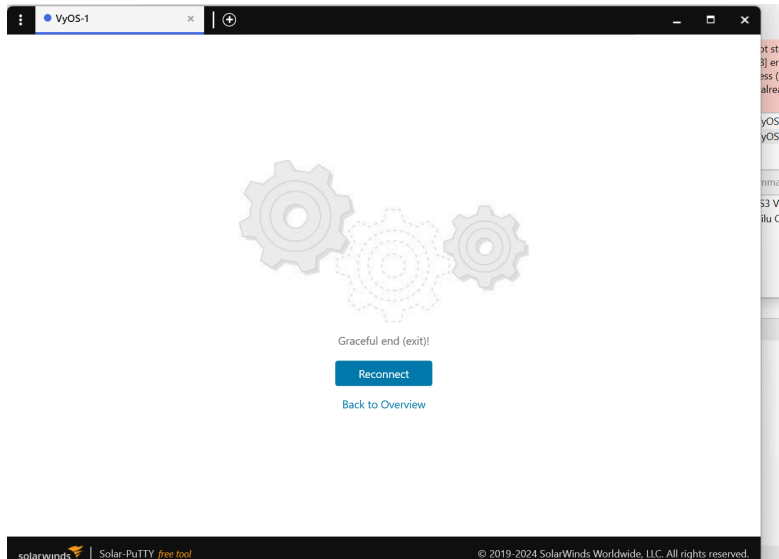
First root cause identify cheyyi. Tarvatha GNS3 VM, QEMU/KVM, VyOS appliance/template configuration lo em check cheyyalo exact steps lo cheppu. Already configured settings ni unnecessarily change cheyyakunda, safest fix first suggest cheyyi. Finally VyOS successfully boot ayindho verify cheyyadaniki em output/screen kanipinchalo kuda cheppu.

27) Bro, nenu GNS3 VM lo VyOS start cheyyadaniki try chesthunna but error vastundhi. Nenu screenshots attach chesthunna, vatini analyze chesi actual problem ekkada undho identify cheyyi. GNS3 server configuration lo KVM-related issue undha, QEMU configuration issue undha check cheyyi.

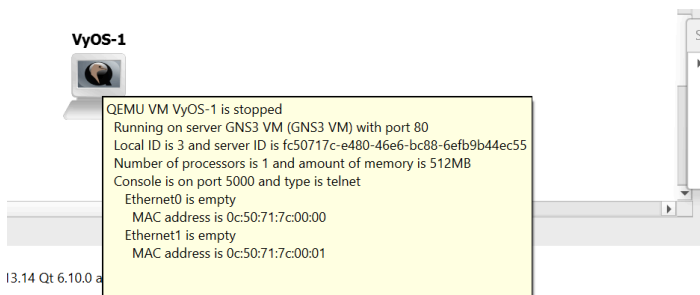


- 28)
- 29) Screenshots lo kanipinche server configuration and paths ni carefully verify chesi, correct config file edi ani identify cheyyi. Guess cheyyakunda, running GNS3 server actually use cheshtunna configuration file ni determine chesi, KVM acceleration error ni resolve cheyyadaniki exact commands and step-by-step procedure ivvu.
- 30) Bro, previous KVM error fix chesaka ippudu different error vastundhi. Screenshot ni check chesi previous error tho compare cheyyi and ippudu `qemu/nodes/.../start` not found error enduku vastundho cheppu. GNS3 VM running and connected undhi. Guess cheyyakunda exact root cause identify chesi safest step-by-step fix cheppu.

cheyyalo step-by-step cheppu. QEMU settings lo boot priority, console type, ISO/CD-ROM configuration and other relevant settings em undalo verify cheyyadaniki guide cheyyi. Random settings change cheyyakunda, first check cheyyalsina settings matrame cheppu.



- 37) 38) Telnet console disconnected because the VyOS QEMU process exited.....Y it hpnd soo brohhh
- 39) Could not start Telnet QEMU console [Errno 98] error while attempting to bind on address ('0.0.0.0', 5000): [errno 98] address already in use...showing this error.....Y and how to resolve this



empting to bind on address ('0.0.0.0', 5000): [errno 98] address already in use

- 40)
- 41) Given this prompt to Claude : GNS3 2.2.61 on Windows, official GNS3 VM in VirtualBox. VyOS rolling vyos-2026.09.09-0029-rolling-generic-amd64 attached to a manually created QEMU template..... GNS3 VM KVM support is False because nested VirtualBox doesn't expose /dev/kvm. We found the actual GNS3 server command uses --config /opt/gns3/server/gns3_server.conf. Added [Qemu] enable_kvm = false and require_kvm = false there and rebooted. KVM error disappeared. VyOS node now starts. Port 5000 was initially occupied; sudo fuser

-k 5000/tcp fixed it. VyOS console is Telnet on port 5000. ISO is attached in the CD/DVD tab and boot priority was set to CD/DVD-ROM or HDD. Current problem: Solar-PuTTY connects successfully but remains completely blank; no vyos login, no boot menu, no output. Node appears running/green in GNS3.

42) Ping fail ayyindi brohhh. Network topology correct laga undi but PC1 nunchi Router1 reach avvattledu. First possible causes list cheyyakunda, systematic gaa first check cheyyalsina one thing cheppu.

43) Interface up undi but IP communication work avvattledu. IP address, subnet mask and interface status verify cheyyadaniki VyOS and VPCS commands ivvu.

44) Brohhh VyOS lo eth0, eth1 ante exact gaa enti? Router ki interfaces ki relation enti? SNMP lo interface metrics collect chestunnappudu idi enduku important?

45) GNS3 lo devices add chesanu but links connect cheyyadam lo confusion undi..... PC and router interfaces ni correct gaa ela connect cheyyali? Ethernet0/eth0 difference kuda cheppu.

46) Brohhh..... ippudu ping work avutundi. Next SNMP configure cheddham. VyOS router lo SNMPv2c enable cheyyadaniki exact configuration steps ivvu and each command enduku use chestunnamo explain cheyyi.

47)

For Phase 2, implement the SNMP collector foundation using the existing metric-source contract/interface.

Before writing the implementation:

1. Explain exactly what SNMP data we will collect.
2. Explain the OIDs we will use, especially:
 - ifHCInOctets
 - ifHCOctets
 - ifSpeed
 - ifOperStatus
 - ifInErrors
 - ifOutErrors
 - interface description/name
3. Explain how interface discovery will work.
4. Explain the difference between raw SNMP counters and the bandwidth/utilization metrics we will calculate later.
5. Make the SNMP configuration come from configuration/environment variables. Do not hardcode IP addresses, community strings, usernames, passwords, or other credentials.
6. Design it so SNMPv2c can be used initially and SNMPv3 can be added cleanly.

7. Do not require access to the IITGN router yet. The implementation must be testable without a live device using mocks/unit tests.
8. Do not implement gRPC, PostgreSQL, FastAPI, Grafana, or alerts yet.
9. Add tests for the SNMP parsing/collection layer using mocked SNMP responses.

At the end:

- show all files created/modified
- explain each file
- give exact Windows PowerShell commands to install/run/test
- run/describe the tests and expected result

47) “Bro, mana Network Telemetry project lo first 3 phases already complete ga unnayi. Ippudu remaining system ela work avuthundho first explain cheyyi — collector nunchi telemetry gRPC dwara backend ki ela velthundhi, backend PostgreSQL lo ela store chesthundhi, alerts ela generate avuthayi, and Grafana lo data ela visualize avuthundho clear ga explain cheyyi. Real SNMP devices primary source ga, simulator development/testing kosam secondary source ga ela use cheyyalo kuda explain cheyyi.

48) SNMPv2c enable chesam. But asalu collector ki router metrics ela vastayo complete flow explain cheyyava? SNMP Manager, Agent, OID, MIB roles enti?

49) Bro, mana Network Telemetry project lo SNMP configure chesthunnappudu “Unknown token” error vachina case ni solve cheyyali. Error message/log and current snmpd.conf configuration ni carefully analyze chesi, exact ga which line/token invalid or misplaced undo identify cheyyi. Existing SNMP setup ni unnecessarily change cheyyaku. Correct configuration syntax ni ivvu, exact commands to fix it cheppu, and finally snmpd successfully running undho and SNMP response vastundho verify cheyyadaniki commands kuda ivvu

50) Broohhh ubutu thhoooo kuda chese Approachs and cmd cheppuu?????????

51) et-iitgn-vm@set-iitgn-vm:~\$ ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
default qlen 1000
```

```
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

```
inet 127.0.0.1/8 scope host lo
```

```
valid_lft forever preferred_lft forever
```

```
inet6 ::1/128 scope host
```

```
valid_lft forever preferred_lft forever
```

```
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state
UP group default qlen 1000
```

```
link/ether 08:00:27:58:33:64 brd ff:ff:ff:ff:ff:ff
```

```
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
```

```
valid_lft 86295sec preferred_lft 86295sec
```

```

    inet6 fe80::d49:96f4:4c82:fc40/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group
default qlen 1000
    link/ether f2:65:b0:ce:1a:d0 brd ff:ff:ff:ff:ff:ff
4: s4: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
qlen 1000
    link/ether 86:8a:f3:e2:7a:4a brd ff:ff:ff:ff:ff:ff
5: s1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
qlen 1000
    link/ether ba:74:17:37:c9:45 brd ff:ff:ff:ff:ff:ff
6: s3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
qlen 1000
    link/ether be:73:44:58:26:44 brd ff:ff:ff:ff:ff:ff
7: s2: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
qlen 1000
    link/ether ba:e7:e5:d3:ed:40 brd ff:ff:ff:ff:ff:ff
.....chudu brojhhhh ip add

```

52)Broohhhh ... by mistake sudo nano /etc/snmp/snmpd.conf ee cmd run chesaa

....dhnatloo data motham delete ayyipoooyindhii ,,,,by mistake

53)set-iitgn-vm@set-iitgn-vm:~\$ sudo ls -l /etc/snmp/snmpd.conf*

-rw----- 1 root root 3004 Sep 13 04:48 /etc/snmp/snmpd.conf

-rw----- 1 root root 3004 Sep 13 04:48 /etc/snmp/snmpd.conf.backup

54)set-iitgn-vm@set-iitgn-vm:~\$ snmpwalk -v2c -c telemetry localhost

1.3.6.1.2.1.31.1.1.1.1

55)Timeout: No Response from localhost

.....brohhh chudu

56)I'm at this stage in the GNS3 VM setup. The terminal shows eth0, eth1, and eth2

as UP, while virbr0 and docker0 are DOWN. This is what I'm currently seeing.

Continue from here based on our previous setup and tell me what to do next. Keep the instructions short and step-by-step.

57)Broohhhh ...can you make as separte brohhh ..then i have like simualtion, 2 as

GNS3, AND 3 AS Ubuntu VM.that maskes it more powerso cna you make it as separte file then i iwll run the code by not chnage manully

58)Bro, SNMP manually verify ayyindi..... Ippudu Python Telemetry Collector

build cheddham. First code rayakunda collector architecture/modules ela divide cheyyali ani design cheyyi.

59) Given to calude: Bro, before making any changes, I want to understand our current Network Telemetry project properly. Please inspect the existing implementation and give me a detailed assessment only — don't install GNS3/EVE-NG, modify Docker or any existing component, delete anything, or create new files yet. First explain the current architecture and complete data flow from Simulator/SNMP → Collector → Processing → gRPC → Backend → PostgreSQL → Grafana, and identify which files handle each stage. Then explain our custom simulator in detail — how router-01, router-02, switch-01 and switch-02 are created, how interfaces and telemetry are generated, whether cumulative counters are used, and how bandwidth/utilization are calculated. Also review the existing SNMP collector and explain which SNMP library/version, OIDs, device configuration format and authentication/community mechanism are currently being used. Finally, assess how suitable the current architecture is for integrating real virtual routers/switches through GNS3 or EVE-NG, including the required configuration/code changes, installation requirements, risks, and testing steps.

For this response, I only want the assessment and explanation. Don't implement anything. I'll review your assessment first and explicitly tell you when to proceed with implementation.

60) Collector lo MetricSource interface use cheddham anukuntunna. Simulator source and SNMP source rendu same interface implement chesthe architecture ela untundi. Why is this useful?

61) Okay brohhh, first MetricSource interface and SNMP implementation create cheddham. Production-quality but simple Python code ivvu, unnecessary frameworks use cheyyaku.

62) Collector periodic polling ela implement cheyyali? Suppose poll interval 5 seconds. Actual elapsed time measure chesi next sample tho bandwidth calculate cheyyadaniki correct design cheppu

63) Bro, first poll ki previous counter undadu kadha. First sample ni ela handle cheyyali? Bandwidth calculate cheyyakunda baseline create cheyyadam correct aa?

64) Counter previous value kanna current value thakkuva vachindi. Idi negative bandwidth istundi. Counter reset/wraparound possibility ni collector lo ela handle cheyyali?

65) Collector code run chesthe ModuleNotFoundError vachindi. Naa environment lo dependencies missing anipistunnayi. First environment verify cheyyadaniki commands ivvu, then only required package install cheddham

- 66) Brohhh SNMP response vastundi but interface metrics empty ga
vastunnayi..... OID wrong aa, interface index issue aa, permissions aa.....
Actual output ni inspect chesi root cause identify cheyyadaniki debugging
approach cheppu.
- 67) ppudu raw counters successfully collect avuthunnayi. Next collector lo bandwidth
and utilization calculation add cheyyali. Formula, units and edge cases explain
chesi code update cheyyi.
- 68) One confusion bro..... poll interval exactly 5 seconds ani assume chesthe
bandwidth inaccurate avvacha..... Actual timestamps madhya elapsed seconds
use cheyyadam enduku important?
- 69) set-iitgn-vm@set-iitgn-vm:~\$ sudo tcpdump -ni enp0s3 udp port 161
[sudo] password for set-iitgn-vm:
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
- 70) Collector config: config/collector-ubuntu.yaml
Device config: config/devices-ubuntu.yaml
2026-09-13 06:07:20,247 INFO collector.main: Collector starting: 1 device(s) configured,
poll interval=10.0s
2026-09-13 06:07:25,055 WARNING collector.snmp.snmp_source: Device ubuntu-01
(10.0.2.15) unreachable: 10.0.2.15:161 GET 1.3.6.1.2.1.1.3.0 failed: No SNMP response
received before timeout
2026-09-13 06:07:42,977 WARNING collector.snmp.snmp_source: Device ubuntu-01
(10.0.2.15) unreachable: 10.0.2.15:161 GET 1.3.6.1.2.1.1.3.0 failed: No SNMP response
received before timeout.....
.....fix chyavva brohhhh
- 71) set-iitgn-vm@set-iitgn-vm:~\$ sudo grep -nE "agentAddress|agentaddress|rocommunity"
/etc/snmp/snmpd.conf
41:# agentaddress: The IP address and port number that the agent will listen on.
49:agentaddress 0.0.0.0:161
67:# rocommunity: a SNMPv1/SNMPv2c read-only access community name
72:#rocommunity6 public default -V systemonly
73:rocommunity telemetry default
- 72) grep -nE "agentAddress|agentaddress|rocommunity" /etc/snmp/snmpd.conf
- 73) >> asyncio.run(main())
>> '@ | & "..\venv\Scripts\python.exe"
Traceback (most recent call last):
File "<stdin>", line 17, in <module>
File "C:\Python314\Lib\asyncio\runners.py", line 205, in run
return runner.run(main)
~~~~~  
File "C:\Python314\Lib\asyncio\runners.py", line 128, in run

```

return self._loop.run_until_complete(task)
~~~~~^
File "C:\Python314\Lib\asyncio\base_events.py", line 719, in run_until_complete
 return future.result()
~~~~~^
File "<stdin>", line 6, in main
File
"C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Fram
ework-main\Code\collector\snmp\snmp_source.py", line 76, in poll_device
    raw_uptime = await transport.get(OID_SYS_UPTIME)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Fram
ework-main\Code\collector\snmp\transport.py", line 108, in get
    ObjectType(ObjectIdentity(oid)),
    ^^^

```

NameError: name 'oid' is not defined. Did you mean: 'id'?.....fx this error brohhh.....

- 74) Bro, GNS3 router setup sometimes difficult avuthundi. Mana architecture test cheyyadaniki custom simulator kuda add cheddham. Simulator real router counters laga realistic data generate cheyyali..... Design suggest cheyyi.
- 75) Simulator random bandwidth values generate cheyyadam enough aa?.....  
Better realistic cumulative RX/TX counters ..... generate chesi collector same calculation pipeline use cheyyadam ela untundi?
- 76) MetricSource interface use chestunnam kadha. .... Simulator and SNMP source interchangeable gaa work cheyyali..... Collector main logic change cheyyakunda source selection ela implement cheyyali?
- 77) Brohhh simulator data work avutundi, SNMP data kuda work avutundi.....  
But rendu sources different field names return chestunnayi..... Common telemetry model create chesi normalize cheyyadam correct aa?
- 78) PS C:\Users\saila> ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:

```

Request timed out.
Request timed out.
Request timed out.
Request timed out.

```

- 79) Ping statistics for 192.168.56.10:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
PS C:\Users\saila>
- 80) OK brohhh now our main moto is instead of custom simulator i wanna integrate a virtual simulator using gns3 cisco 1000 smnt

```

81) collector-1 | 2026-09-12 23:32:41,432 WARNING collector.grpc_client.client:
    gRPC send attempt 1/3 failed (StatusCode.UNAVAILABLE); retrying in 2.0s

    collector-1 | 2026-09-12 23:32:43,435 WARNING collector.grpc_client.client:
    gRPC send attempt 2/3 failed (StatusCode.UNAVAILABLE); retrying in 4.0s

    collector-1 | 2026-09-12 23:32:47,440 WARNING collector.grpc_client.client:
    gRPC send attempt 3/3 failed (StatusCode.UNAVAILABLE); retrying in 8.0s

    collector-1 | 2026-09-12 23:32:55,450 ERROR collector.grpc_client.client:
    Backend unavailable after 3 attempts; dropping this batch (13 records): failed
    to connect to all addresses; last error: UNKNOWN: ipv4:127.0.0.1:50051: Failed
    to connect to remote host: connect: Connection refused (111)

```

.....Fix it Broohhh

- 82) Collector ippudu metrics calculate chestundi. Next backend ki send cheyyali.  
REST already possible unna situation lo project lo gRPC enduku use  
chestunnam..... Collector..... Backend communication ki exact reason  
explain cheyyi.
- 83) Bro, gRPC .proto file asalu enduku..... TelemetryBatch, MetricSample,  
SendTelemetry() laga mana project ki suitable proto design create cheyyi and each  
field explain cheyyi. ....
- 84).proto create chesanu but generated Python files ela create avuthayo confusion  
undi.....Windows/PowerShell environment ki exact commands ivvu and  
generated files ekkada undalo cheppu.
- 85) gRPC client run chesthe connection refused vachindi brohhh..... Collector  
correct port ki connect avuthundha, backend running aa, proto service name match  
avutunda ani first verify cheddham. ....One step at a time guide cheyyi.
- 86) gRPC request ippudu backend ki reach avutundi..... But telemetry batch lo  
timestamp and device ID missing gaa backend lo kanipistunnayi..... Proto  
serialization/deserialization ni inspect chesi issue identify cheyyi.
- 87) Bro, backend temporarily stop chesthe collector crash avvakudadhu. .... Retry  
with exponential backoff implement cheyyali. .... Retry behavior and  
maximum retry strategy simple gaa design cheyyi.
- 88) Ippudu FastAPI backend build cheddham brohhh. Backend responsibilities exact  
gaa define chesi project folder structure suggest cheyyi.
- 89) Router ..... Service ..... Repository architecture use chestunnam. Ee three layers  
lo exact responsibility enti? Telemetry request flow example tho explain cheyyi.
- 90) FastAPI backend lo gRPC telemetry receive cheyyadaniki service implement  
cheyyi, and simultaneously historical metrics kosam REST endpoints expose  
cheyyadaniki structure create cheyyi.

- 91) Backend telemetry receive chestundi but PostgreSQL ki insert avvattledu brohhh.  
First database connection verify cheyyala or repository code check cheyyala?  
Don't guess — debugging sequence ivvu.
- 92) PostgreSQL connection work ayyindi but duplicate telemetry records vastunnayi  
because collector retry chestundi. Duplicate handling/idempotency design suggest  
cheyyi without breaking current architecture.
- 93) Bro, PostgreSQL schema final cheddham. Devices, telemetry metrics and alerts ki  
tables create cheyyi. Timestamp + device based queries fast gaa undadaniki useful  
indexes kuda explain cheyyi.
- 94) Database lo telemetry records successfully store avutunnayi. Ippudu Grafana  
connect cheddham. PostgreSQL datasource configure chesi bandwidth,  
latency and utilization panels create cheyyadaniki exact steps ivvu.
- 95) PS

```
C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Framework-main\Code> docker compose up --build postgres backend grafana
```

error while interpolating

```
services.backend.environment.POSTGRES_PASSWORD: required variable POSTGRES_PASSWORD is missing a value: POSTGRES_PASSWORD must be set in .env
```

error while interpolating

```
services.grafana.environment.GF_SECURITY_ADMIN_PASSWORD: required variable GRAFANA_ADMIN_PASSWORD is missing a value: GRAFANA_ADMIN_PASSWORD must be set in .env
```

error while interpolating

```
services.grafana.environment.GF_TELEMETRY_DB_PASSWORD: required variable POSTGRES_PASSWORD is missing a value: POSTGRES_PASSWORD must be set in .env
```

error while interpolating

```
services.postgres.environment.POSTGRES_PASSWORD: required variable POSTGRES_PASSWORD is missing a value: POSTGRES_PASSWORD must be set in .env.....brohhhh see this
```

```

✓ Image code-backend      Built
3.7s
✓ Network code_default    Created
0.0s
✓ Volume code_postgres_data Created
0.0s
✓ Volume code_grafana_data Created
0.0s
✓ Container code-postgres-1 Created
0.1s
✓ Container code-backend-1 Created
0.6s
✓ Container code-grafana-1 Created
0.1s

```

Attaching to backend-1, grafana-1, postgres-1

Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint code-postgres-1

(b75f6dcd4b70a5a8493b25a5b397b58319c2356482265d55252a8c5aa06fc143):

Bind for 0.0.0.0:5432 failed: port is already allocated

What's next:

Debug this Compose error with Gordon → docker ai "help me fix this compose error".....broohh see this

97)Grafana datasource connected but panel lo “No data” ani vastundi brohhh.

Database lo records unnayi. Query, timestamp format and Grafana time range ni systematic gaa verify cheyyadaniki one-by-one debugging cheyyi.

98) Bro, one thing I'm confused about in our alert system: when an alert becomes RESOLVED, is that something we have to manually do, or does our system automatically detect that the problem is gone? For example, if bandwidth crosses the threshold and an ACTIVE alert is generated, what exactly happens in the following polling cycles when the bandwidth comes back to normal? Also, in our simulator demo, if we use demo\_scenario.py to change the traffic back to NORMAL, is that considered manually resolving the alert, or is the script only changing the simulated network condition and the alert resolution is still handled automatically by our system? I want to understand the difference between fixing/recovering the actual condition and changing the alert state.”

99)Brohhh complete pipeline work avutundi: Device ...>SNMP .....>Collector .....>gRPC .....>FastAPI .....>PostgreSQL .....>Grafana. Ippudu sir expected

outcomes tho compare chesi em implement ayyindi, em verify ayyindi, em future scope lo undho separate cheyyi.

100) PS

```
C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Framework-main\Code> .venv\Scripts\Activate.ps1
```

(.venv) PS

```
C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Framework-main\Code> python scripts/run_collector_profile.py ubuntu
```

```
--community telemetry
```

Traceback (most recent call last):

File

"C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Framework-main\Code\scripts\run\_collector\_profile.py", line 15, in <module>

from collector.main import build\_app\_from\_config

File

"C:\Users\Hp\Downloads\Network-Telemetry-Framework-main\Network-Telemetry-Framework-main\Code\collector\main.py", line 22, in <module>

import yaml

ModuleNotFoundError: No module named 'yaml'.....what borhhh ..this

101) alerts/active correct data return chestundi kani Grafana Alerts dashboard empty ga undi. Datasource, SQL query, status value, and provisioning inspect cheyyi.

102) Grafana login Invalid username or password vastundi after changing .env. Existing Docker volume issue explain chesi data-loss warning tho safe reset steps ivvu.

103) “Bro, mana existing custom simulator ni use chesi ee scenarios test cheyyali: NORMAL, HIGH\_BANDWIDTH, HIGH\_LATENCY, PACKET\_LOSS, DEVICE\_DOWN and RECOVERY. Existing code/config ni inspect chesi, already available scenario support ni identify cheyyi and each scenario ni trigger cheyyadaniki exact commands generate cheyyi. Commands mana current project structure and implementation ki match avvali — generic/example commands ivvakunda actual runnable commands ivvu. Each command run chesaka collector/backend lo em happen avvali and expected result enti ani kuda mention cheyyi.”

104) Broohhh .....Can you Provide the me overall cmds to run this Project that makes easy to demonstrate.

